

FINAL PROJECT: BASIC AUDIO EQUALIZER

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Abstract:

An audio equalizer is a circuit that amplifies a specific frequency range in an audio signal. Where the main purpose of the equalizer is to change the frequency of an audio system so that the output sound matches preferences of the user. The purpose of this project is to understand the application of low-pass, band-pass (which is mid), and high-pass filters in audio processing. Which is then used to build an audio amplifier where it will change the input audio frequency by changing its resistance. This amplifier will also allow us to achieve independent control of bass, midrange, and treble frequencies which can be useful for other applications.

Key Functions of the Audio Equalizer:

Some of the main key functions of the audio equalizer is:

- Frequency band separation: Where we divide the audio signal into low, mid, and high-frequency bands.
- Gain control: Where adjusting the amplitude of each frequency band independently using multiple potentiometers.
- Signal conditioning: Improving the audio clarity by cutting specific frequency components.
- Noise reduction: Minimizing unwanted frequency components.
- Signal amplification: Ensuring sufficient output level for further amplification stages.

Applications for Audio Equalizers:

Some well-known applications for audio equalizers are used in:

- Home audio systems and music players where get to adjust the volume when playing a song or video
- Professional sound systems where we can amplify the noise to make it clear and smoother
- Recording studios to adjust the input voice when singing
- Car audio systems to adjust the volume when playing a radio or song

Theory:

Filters are used such that it allows only a certain frequency range to pass while ignoring other ones. For this project active filters are used, which combine the resistors, capacitors, and operational amplifiers.

Low-Pass Filter (LPF):

Low-pass filter is used such that it allows only frequencies below a cutoff frequency to pass while ignoring higher frequencies. It is mainly used for bass control.

Cutoff frequency equation:

$$f_c = \frac{1}{2\pi RC}$$

Where R is the resistor value used in the filter circuit and C is the capacitor value used in the filter.

Working Principle:

- At low frequencies the capacitor offers high impedance allowing the signal to pass.
- At high frequencies the capacitor offers low impedance which ignores the signal.

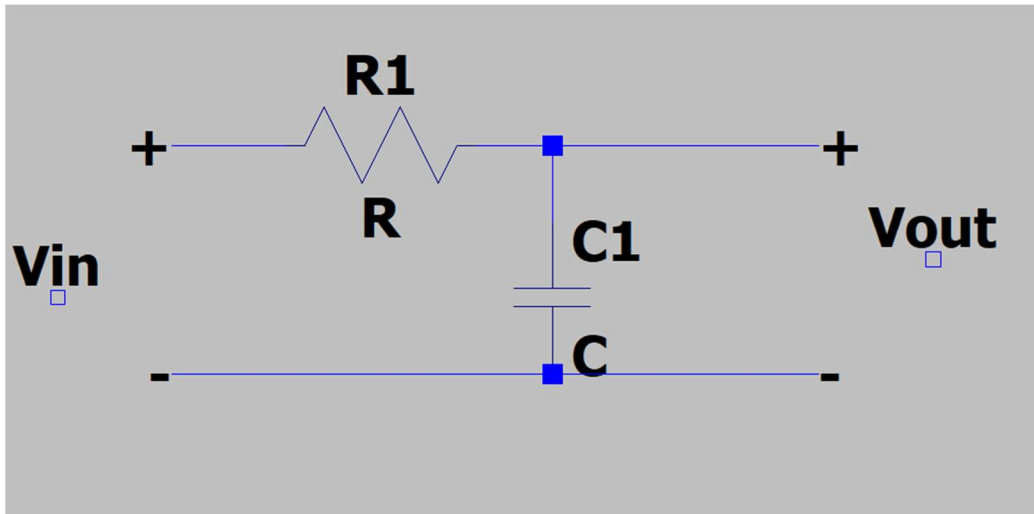


Figure 1: Circuit of Low-Pass Filter.

High-Pass Filter (HPF):

A high-pass filter allows frequencies above the cutoff frequency to pass while ignoring low frequencies. It is used for treble control.

Cutoff frequency equation:

$$f_c = \frac{1}{2\pi RC}$$

Where R is the resistor value used in the filter circuit and C is the capacitor value used in the filter.

Working Principle:

- At low frequency the capacitor blocks the signal.
- At high frequencies the capacitor allows the signal to pass.

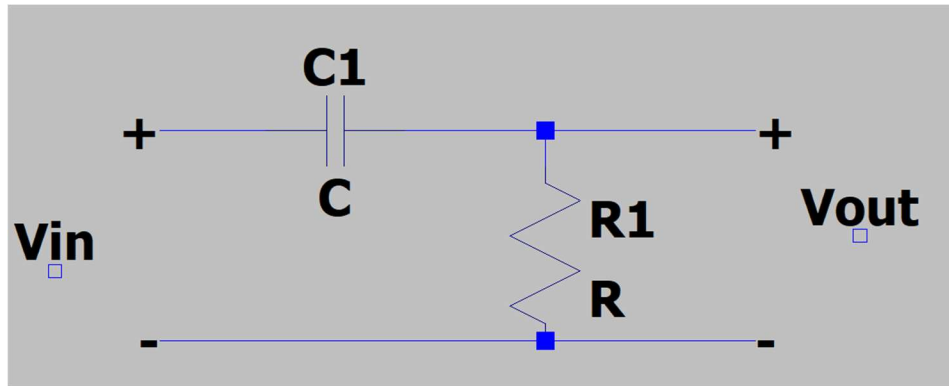


Figure 2: Circuit of High-Pass Filter.

Band-Pass Filter (BPF)

A band-pass filter allows a specific range of frequencies (which is mid-range) to pass while ignoring the frequency outside this range. We use the high pass and low pass filters create the circuit.

Center frequency equation:

$$f_0 = \sqrt{f_L f_H}$$

Where f_L = lower cutoff frequency and f_H = higher cutoff frequency.

Working Principle:

- Combines the filters of low-pass and high-pass filters.
- Used for controlling mid frequencies in audio equalizers.

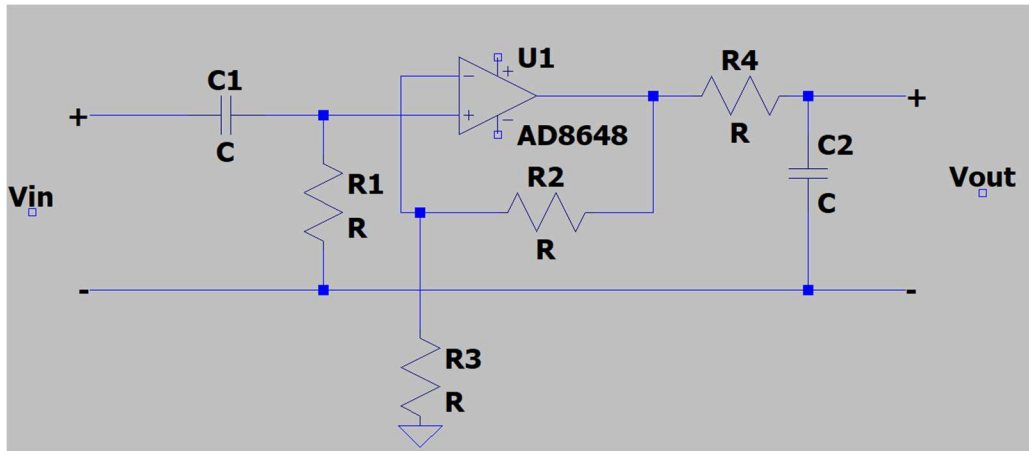


Figure 3: Circuit of Band-Pass Filter.

Operational Amplifier:

An operational amplifier (op-amp) is a high-gain amplifier where it has very high input impedance and low output impedance. In the project the op-amps are used for active filtering and signal amplification. Moreover, the ideal op-amp should have the characteristics of finite open loop gain infinite, input impedance and also zero output impedance.

Inverting Amplifier:

Gain equation:

$$A_v = -\frac{R_f}{R_{in}}$$

Where R_{in} is the input Resistor and R_f is the feedback resistor of the circuit.

Working Principle:

- The input signal is applied to the inverting terminal.
- Output signal is inverted and amplified.
- Used in active filter configurations.

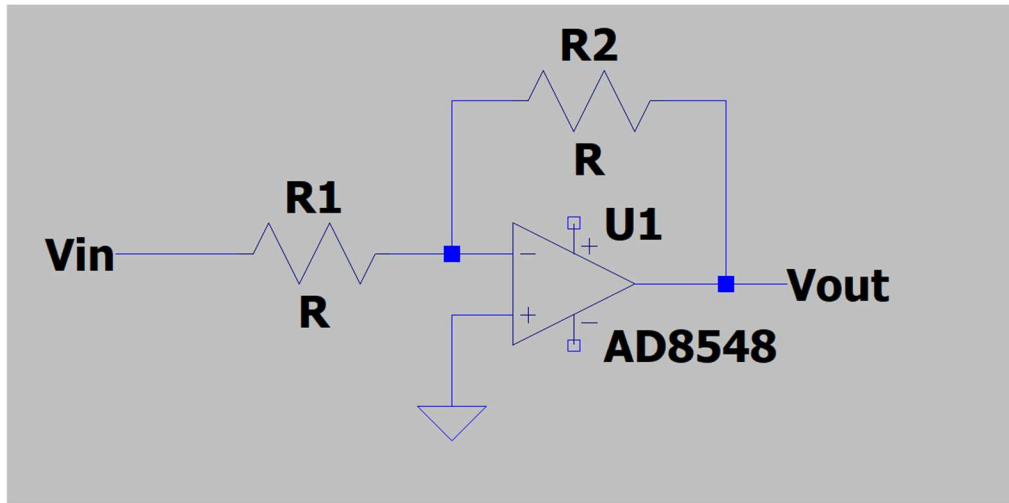


Figure 4: Circuit of Inverting Amplifier.

Non-Inverting Amplifier

Gain equation:

$$A_v = 1 + \frac{R_f}{R_g}$$

Where R_g is the gain resistor and R_f is the feedback resistor of the circuit.

Working Principle:

- Input is applied to the non-inverting terminal.
- Output signal is amplified without phase inversion.
- Used for buffering and gain control in the equalizer.

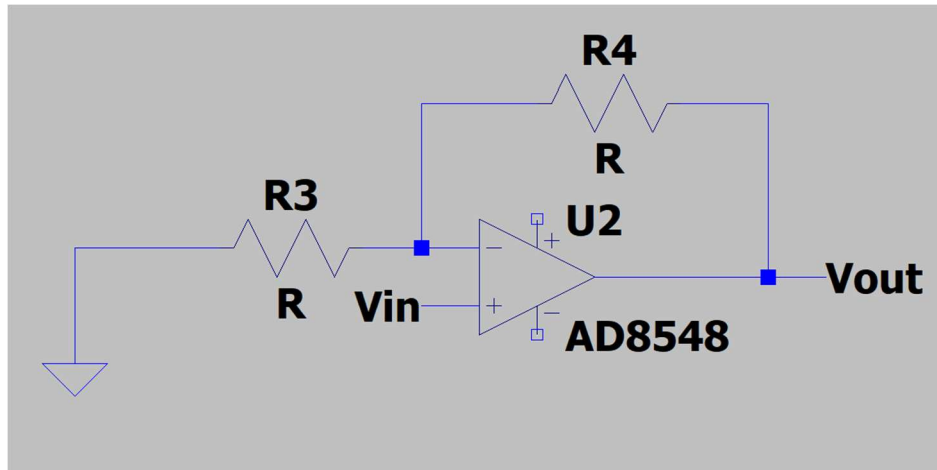


Figure 5: Circuit of Non-inverting Amplifier.

Power Amplifier:

A power amplifier is used to amplify the processed audio signal up to a level where it's sufficient to drive a loudspeaker.

Working Principle:

- It receives low-power audio signals from equalizer output.
- Then increases current and voltage without significantly altering the signal shape.
- Operates in the Class AB mode for better efficiency and low distortion which is commonly used in audio applications.

Output Power of a Power Amplifier Equation

$$P_{out} = \frac{V_{rms}^2}{R_L}$$

Where P_{out} is output power (watts), V_{rms} is RMS output voltage and R_L is load resistance.

RMS Voltage Relation equation

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$

Where it is used to convert peak output voltage to RMS.

Vrms is the RMS voltage of the circuit and Vpeak is the peak voltage of the circuit.

Power Gain Equation

$$A_p = \frac{P_{out}}{P_{in}}$$

Also, in decibels (dB):

$$A_p(dB) = 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right)$$

Where Pout and Pin is the output and input Power of the circuit.

Efficiency of Power Amplifier Equation

$$\eta = \frac{P_{out}}{P_{DC}} \times 100\%$$

Where P_{DC} is the DC power drawn from supply and Pout is the output power.

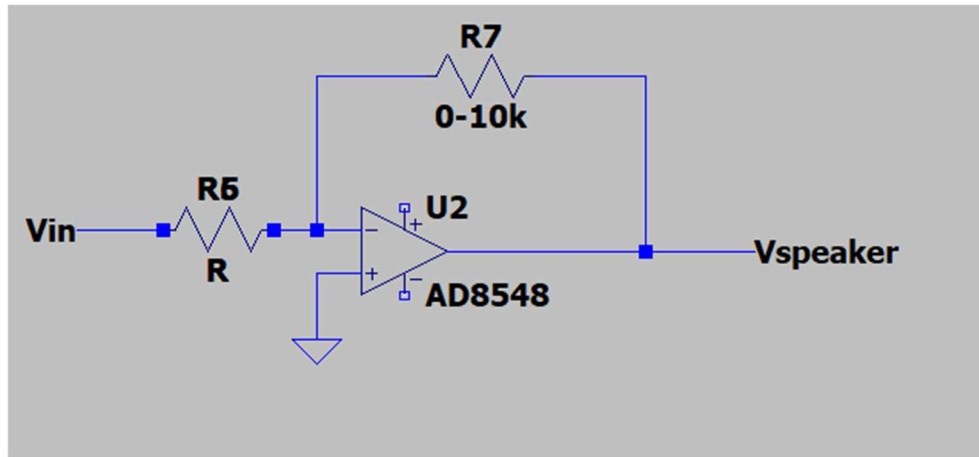


Figure 6: Circuit of Power Amplifier.

Procedure:

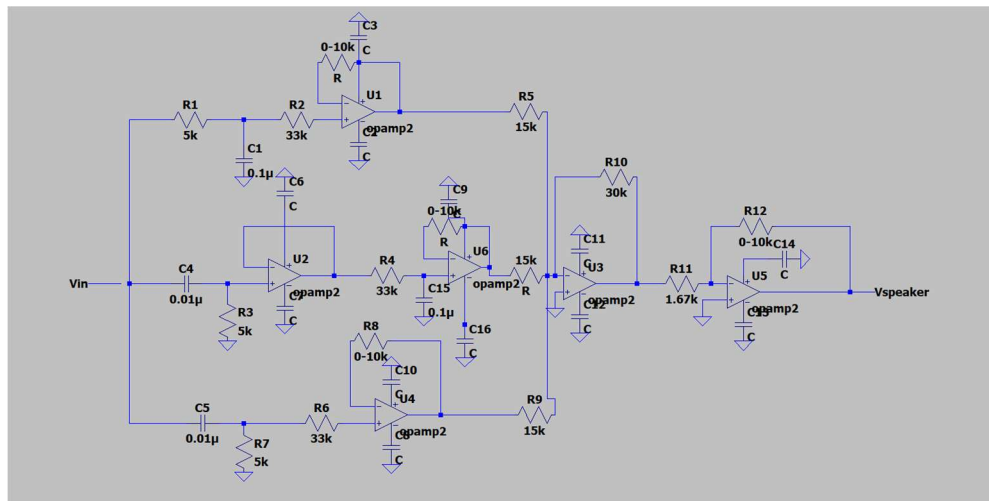


Figure 7: Schematic of the audio equalizer.

Design Parameters Used in the Project:

Filter Design Parameters (Low-Pass, High-Pass, and Mid-Pass):

Low-Pass Filter (Bass Control):

Desired cutoff frequency:

$$f_c = 320 \text{ Hz}$$

Using:

$$f_c = \frac{1}{2\pi RC}$$

Chosen values which can be used in circuit:

- $R = 5 \text{ k}\Omega$
- $C = 0.1 \text{ }\mu\text{F}$

Calculated cutoff frequency (approximate values):

$$f_c \approx 318 \text{ Hz}$$

High-Pass Filter (Treble Control):

Desired cutoff frequency:

$$f_c = 3200 \text{ Hz}$$

Chosen values which can be used in circuit:

- $R = 5 \text{ k}\Omega$
- $C = 0.01 \text{ }\mu\text{F}$

Calculated cutoff frequency (approximate values):

$$f_c \approx 3180 \text{ Hz}$$

Band-Pass Filter (Mid Control):

Lower cutoff frequency: $f_L = 320 \text{ Hz}$

Upper cutoff frequency: $f_H = 3200 \text{ Hz}$

Center frequency (approximate values):

$$f_0 = \sqrt{f_L f_H} \approx 1012 \text{ Hz}$$

The band-pass filter is implemented by a high-pass filter which is followed by a low-pass filter.

Volume Control Circuits Design Parameters:

Each frequency band is provided with an independent volume control circuit for this audio amplifier circuit.

Potentiometer value used: 0-10 k Ω (used in lab)

The output voltage after volume control is given by:

$$V_{out} = V_{in} \times \frac{R_{pot}}{R_{total}}$$

Where V_{in} is the input voltage, R_{pot} is the resistance of potentiometer and R_{total} is total resistance of the circuit.

Minimum Volume Setting

$$R_{wiper} \approx 0$$

$$V_{out(min)} \approx 0$$

This ensures:

$$V_{out} < 15 mV_{RMS}$$

at 100 Hz, 1 kHz, and 10 kHz.

Maximum Volume Setting

$$R_{wiper} \approx R_{pot} = 10 k\Omega$$

$$V_{out(max)} \approx V_{in}$$

Separate volume controls are used for:

- Bass band
- Mid band
- Treble band

Which allows independent adjustment of each frequency range.

Signal Recombination Stage Design Parameters:

Component Values

Input resistor for each band:

$$R_{in1} = R_{in2} = R_{in3} = 10 k\Omega$$

Feedback resistor:

$$R_f = 10 k\Omega$$

Gain Calculation

Using:

$$V_{out} = -R_f \left(\frac{V_L}{R_{in}} + \frac{V_M}{R_{in}} + \frac{V_H}{R_{in}} \right)$$

Also:

$$R_f = R_{in}$$

Each input has unity gain:

$$A_v = -\frac{R_f}{R_{in}} = -1$$

Result

- Each of the frequency bands contribute equally
- Only the summation of signals takes place

However, in the lab we have taken 15k resistor value instead of 10k as we have noticed better outputs in the FRA compared to the 10k ohm resistor value.

Short Description of Operation of the Designed Circuit:

The input audio signal is first buffered using a non-inverting operational amplifier to prevent loading of the source. Then the buffered signal is then divided into three frequency bands using low-pass, band-pass, and high-pass filters designed with the calculated cutoff frequencies of approximately 320 Hz and 3200 Hz.

Each filtered signal is then independently controlled using a 10 kΩ potentiometer which allows the adjustment of bass, midrange, and treble. Then the adjusted signals are recombined using an op-amp summing amplifier with a gain of 1. Finally, the combined signal is then applied to an LM386 power amplifier, which then delivers enough power to drive an 8 ohm resistor speaker.

Result:

Lowpass filter's FRA curves with explanation:

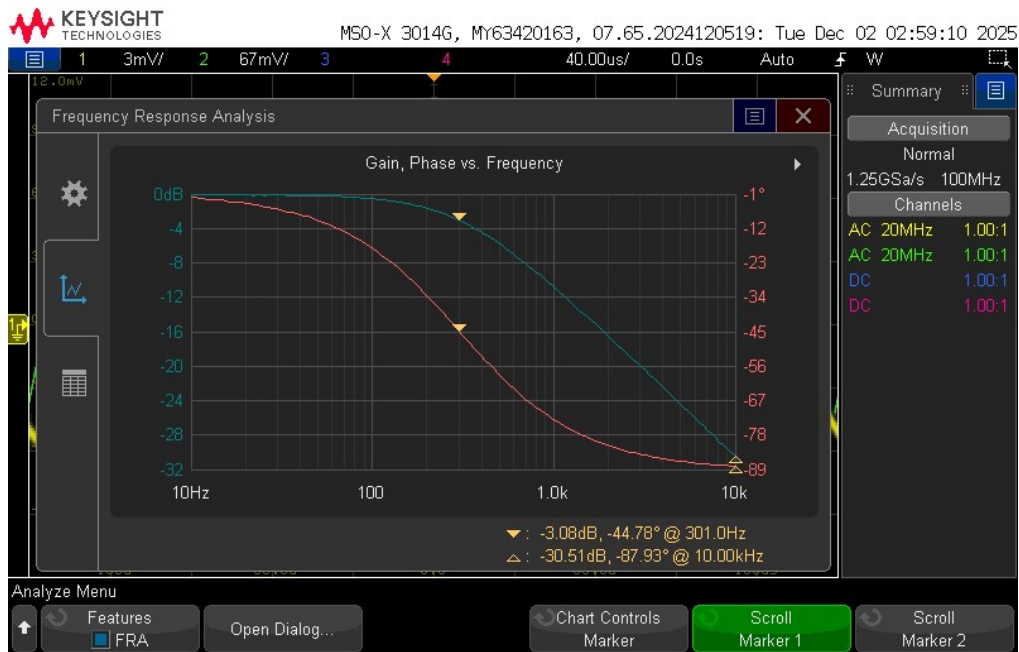


Figure 8: FRA of Low-Pass Filter.

As you can see from the FRA at around -3dB the frequency is 301Hz which is close to the desired frequency of 320Hz with errors of 5.93% which is less than 10%.

High pass filter's FRA curves with explanation:

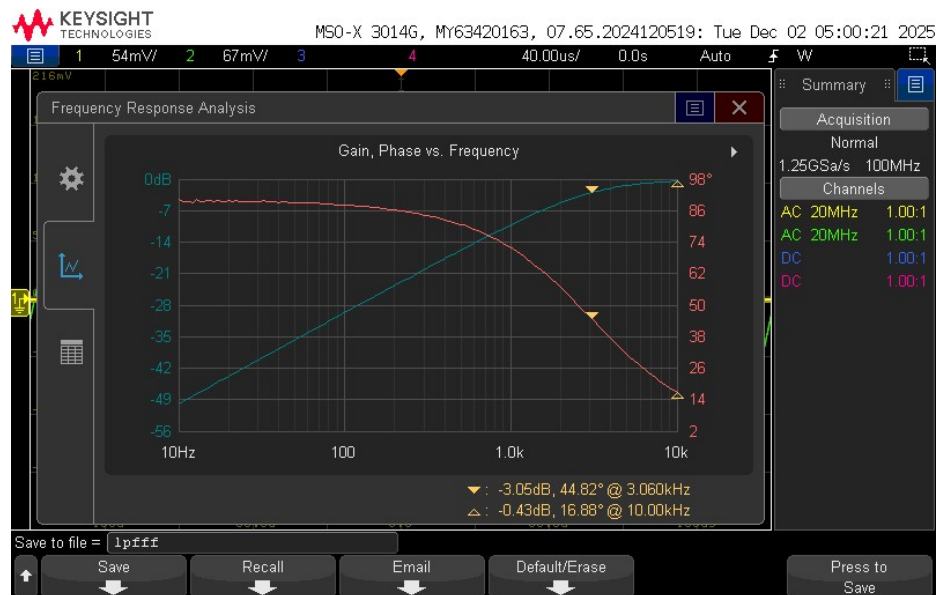


Figure 9: FRA of High-Pass Filter.

As you can see from the FRA at around -3dB the frequency is 3.060 kHz which is close to the desired frequency of 3200 Hz with errors of 4.375%, which is less than 10%.

Mid pass filter's FRA curves with explanation:

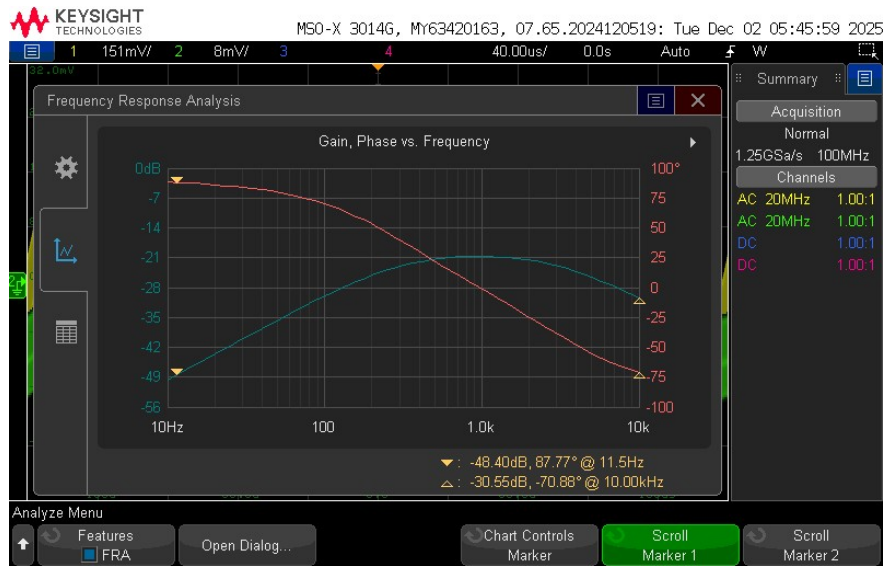


Figure 10: FRA of Mid-Pass Filter.

The measured -3dB points at csv using the FRA:

Lower → 317Hz → Error = 1.5%, Lower Measured Phase → 37.01 deg

Upper → 3002Hz → Error = 6.3%, Upper Measured Phase → 37.5 deg

Which is close to the desired the expected values and is also well within the percentage error.

Plots showing Vamp with all the equalizers turned to minimum setting:

At 100 Hz:

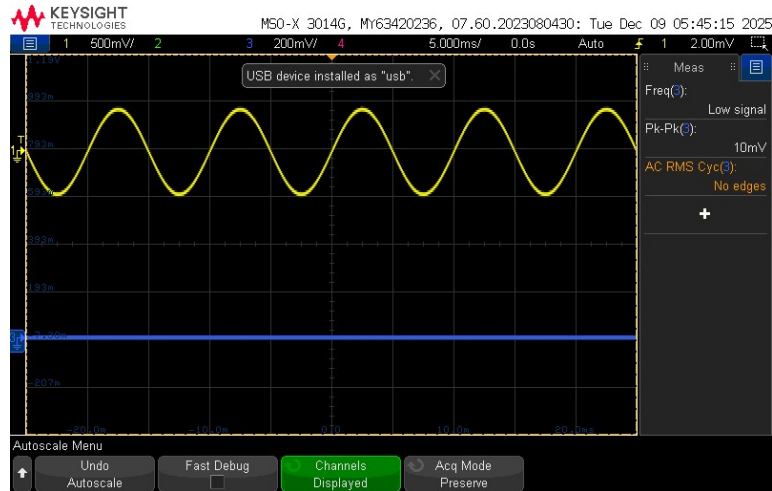


Figure 11: Oscilloscope of equalizer turned to minimum setting at 100Hz.

The plot at 100 Hz shows a sinusoidal waveform with a peak voltage of 10 mV which confirms the reduction of low-frequency (bass) signals when the equalizer is at minimum and is less than 15mV.

At 1000 Hz:

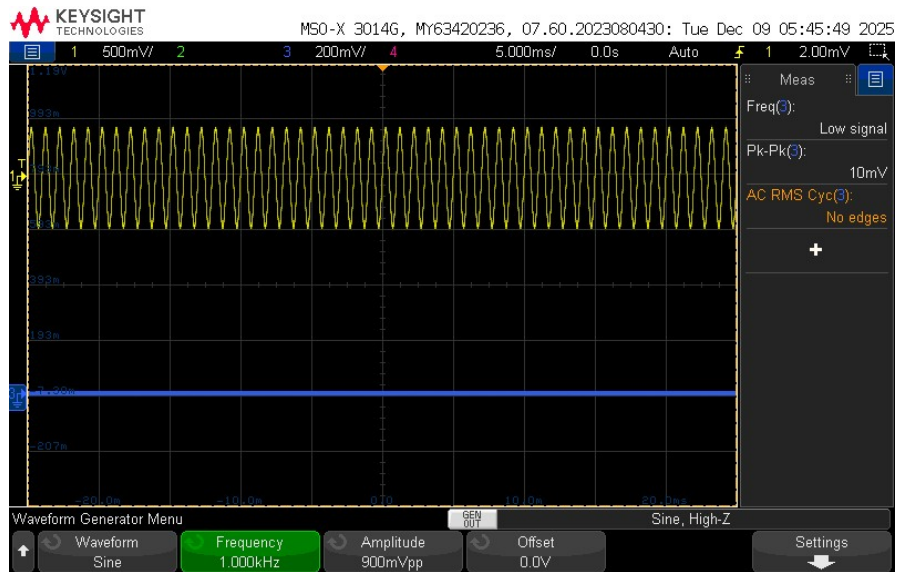


Figure 12: Oscilloscope of equalizer turned to minimum setting at 1000Hz.

The 1kHz plot shows a more sinusoidal waveform in the oscilloscope due to the higher frequency, but the amplitude remains limited to 10 mV which is still less than 15mV.

At 10,000Hz:

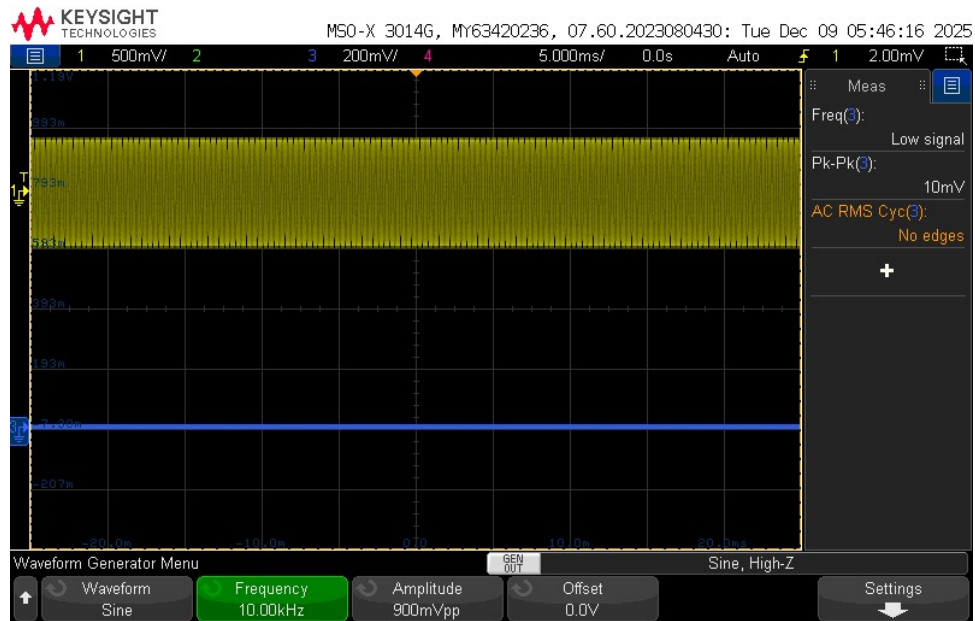


Figure 13: Oscilloscope of equalizer turned to minimum setting at 10,000Hz

The 10kHz plot shows even more sinusoidal waveform than the 1k Hz in the oscilloscope due to the higher frequency being set, but the amplitude remains limited to 10 mV which is still less than 15mV which proves that the equalizer meets the specification for minimum volume operation across the entire audio spectrum which ensures low output noise and minimal signal leakage.

Plots showing Vamp with all the equalizer turned to maximum setting:

At 100 Hz:

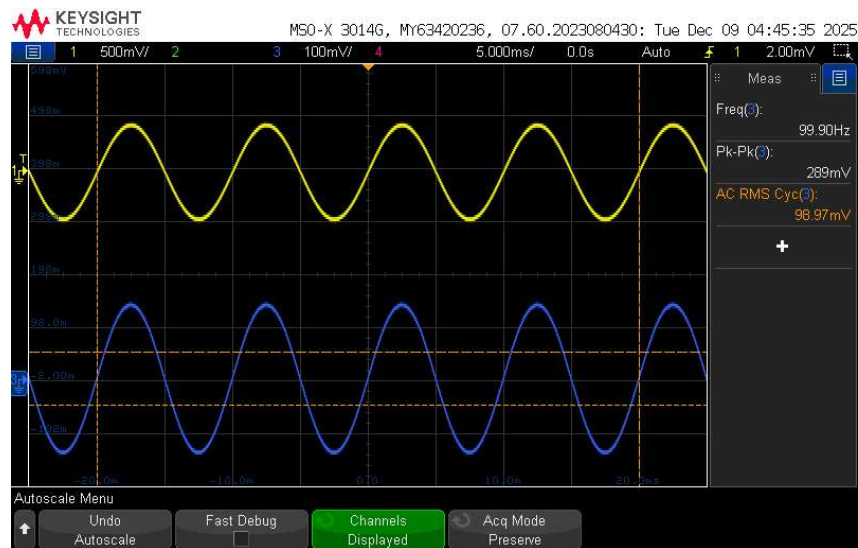


Figure 14: Oscilloscope of equalizer turned to maximum setting at 100 Hz

At 100 Hz the oscilloscope shows the sinusoidal waves with the Vrms of 98.97 mV which are within the acceptable range of 90mV and 110 mV.

At 1000 Hz:

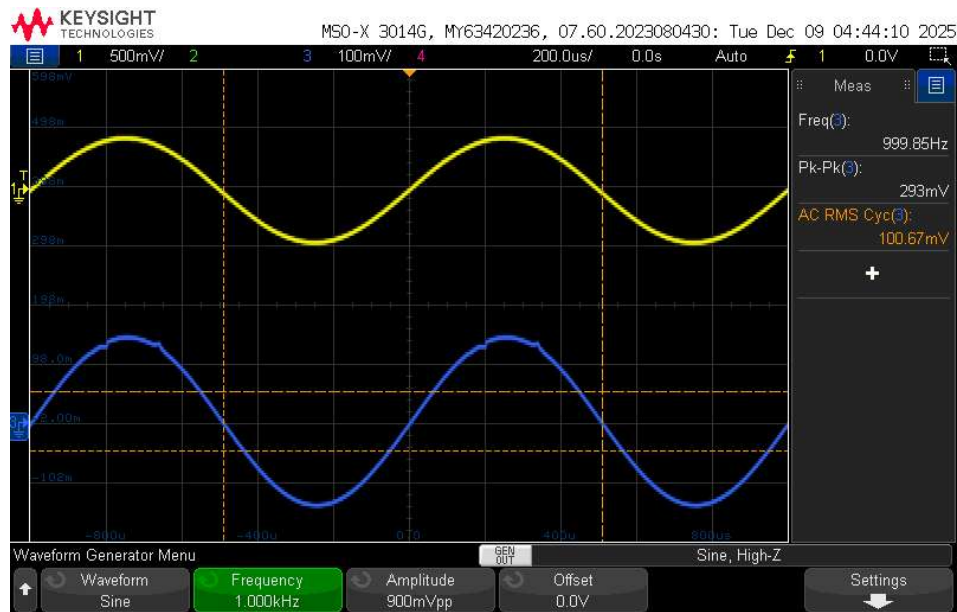


Figure 15: Oscilloscope of equalizer turned to maximum setting at 1000 Hz

At 1 kHz, the output waveform remains in the same sinusoidal shape as 100 Hz with an RMS voltage of 100.67 Hz which are within the acceptable range of 90mV and 110 mV.

At 10,000 Hz:

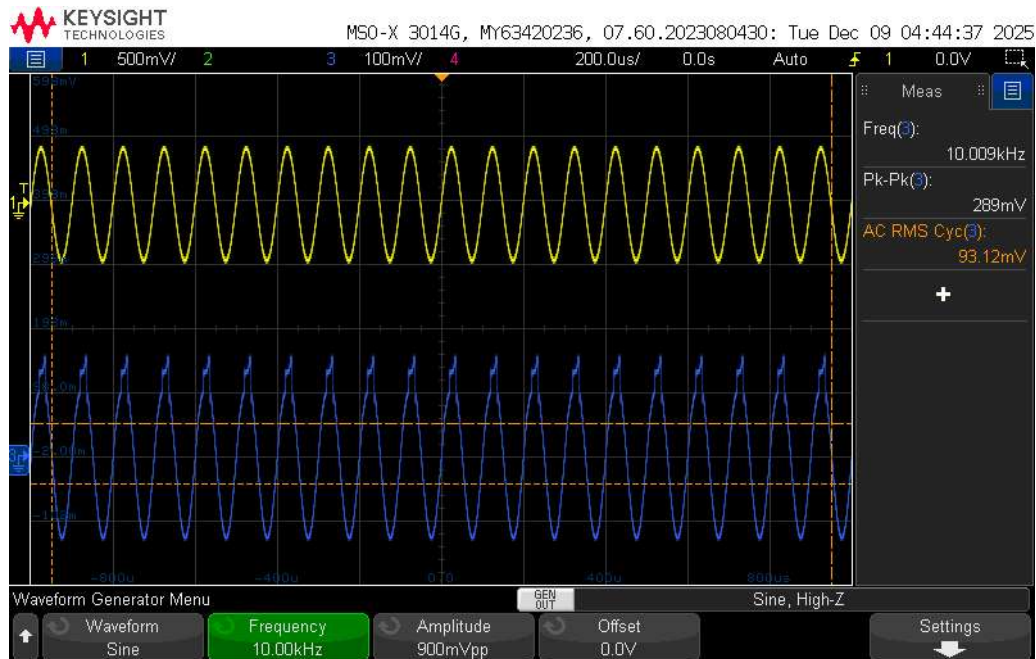


Figure 16: Oscilloscope of equalizer turned to maximum setting at 10,000 Hz

The 10 kHz output shows a high-frequency sinusoidal waveform with a measured RMS voltage of 93.12 mV which are within the acceptable range of 90mV and 110 mV. Which proves that the audio equalizer meets the maximum-volume specification across the entire audio band which ensures a consistent signal amplification before the power amplifier stage.

Plots related to Ripple measurement:



Figure 17: FRA of the final output

Calculating Ripple:

$$V_{\text{ripple(rms)}} = \frac{V_{pp}}{2\sqrt{3}}$$

Where V_{pp} = measured peak-to-peak ripple voltage.

For dB:

$$V_{\text{ripple(rms)}} = 10^{\frac{dB}{20}} \text{ volts}$$

Where dB is the reading at ripple frequency

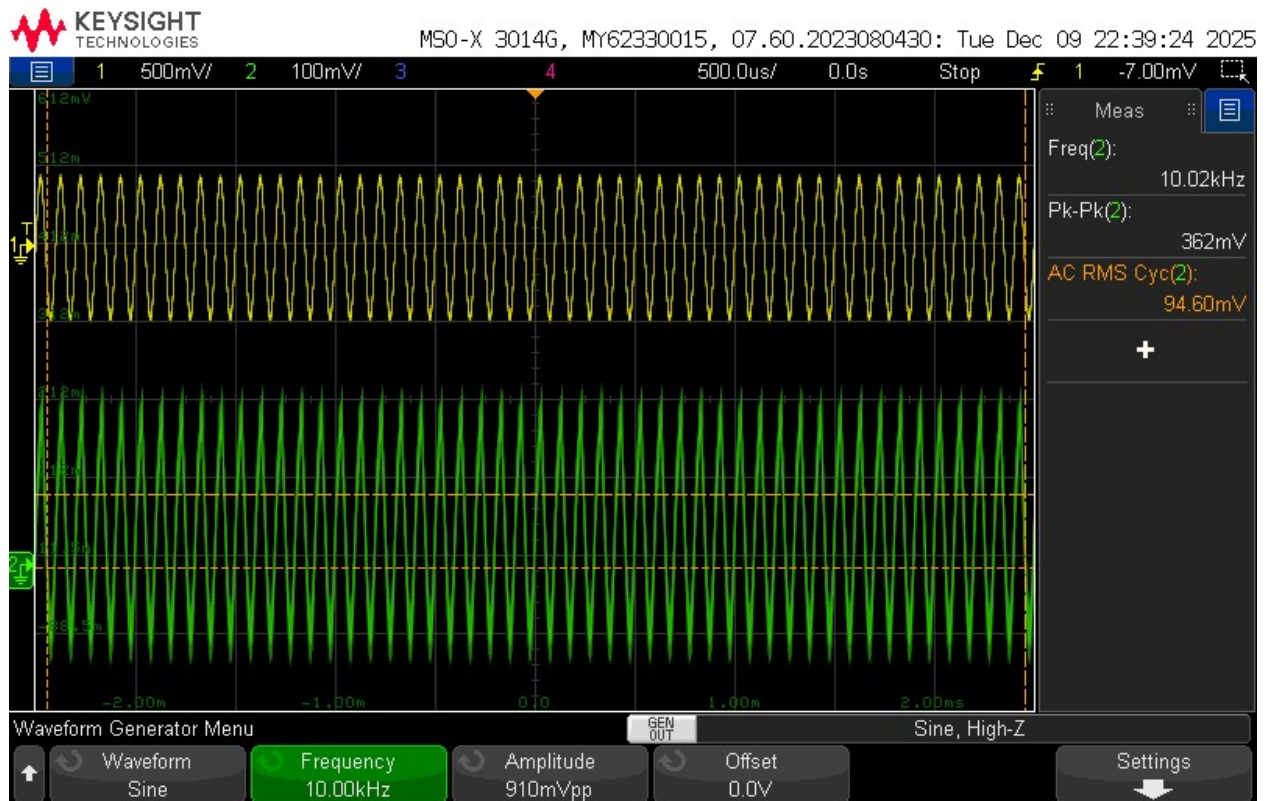


Figure 18: Oscilloscope at 10kHz

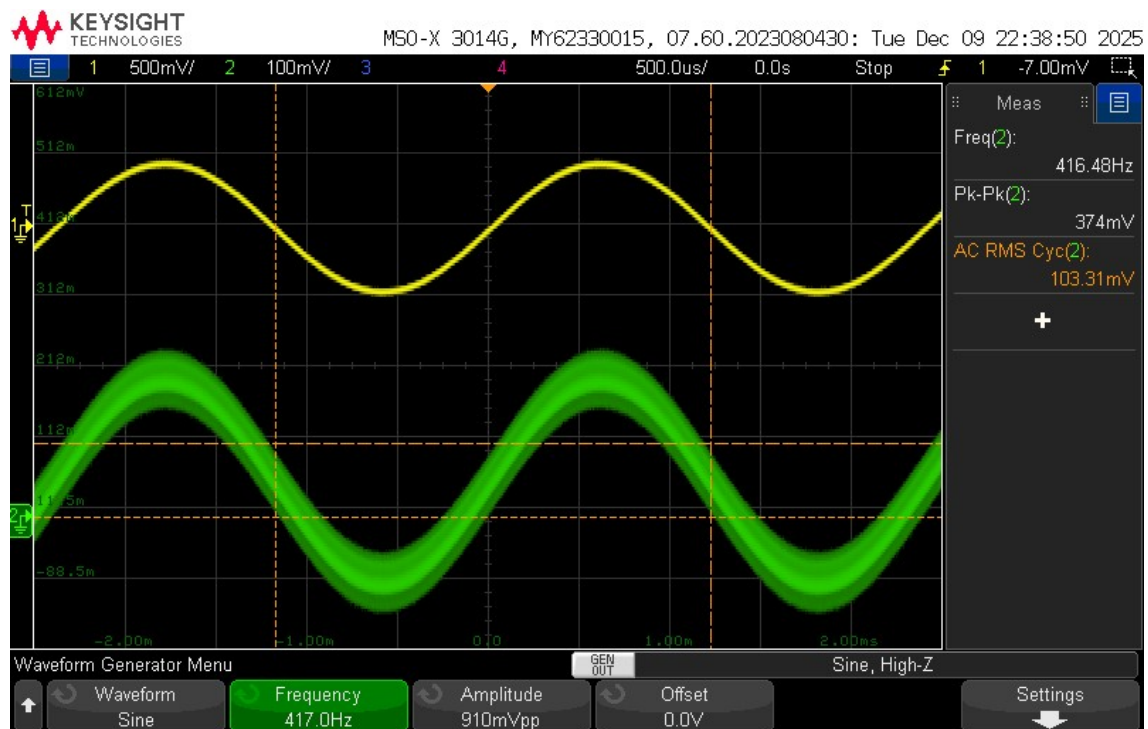


Figure 19: Oscilloscope at 417 Hz

When subtracting their peak to peak the Vrms values comes to which come in the range of the ripple 8.71Vrms which is in the range we expected, however we should keep in mind that the values taken are not with the lowest frequency which is at 100 Hz as a result there will be errors in the final result as seen over here.

Plots showing Power Amplifiers Output:

At 100Hz:

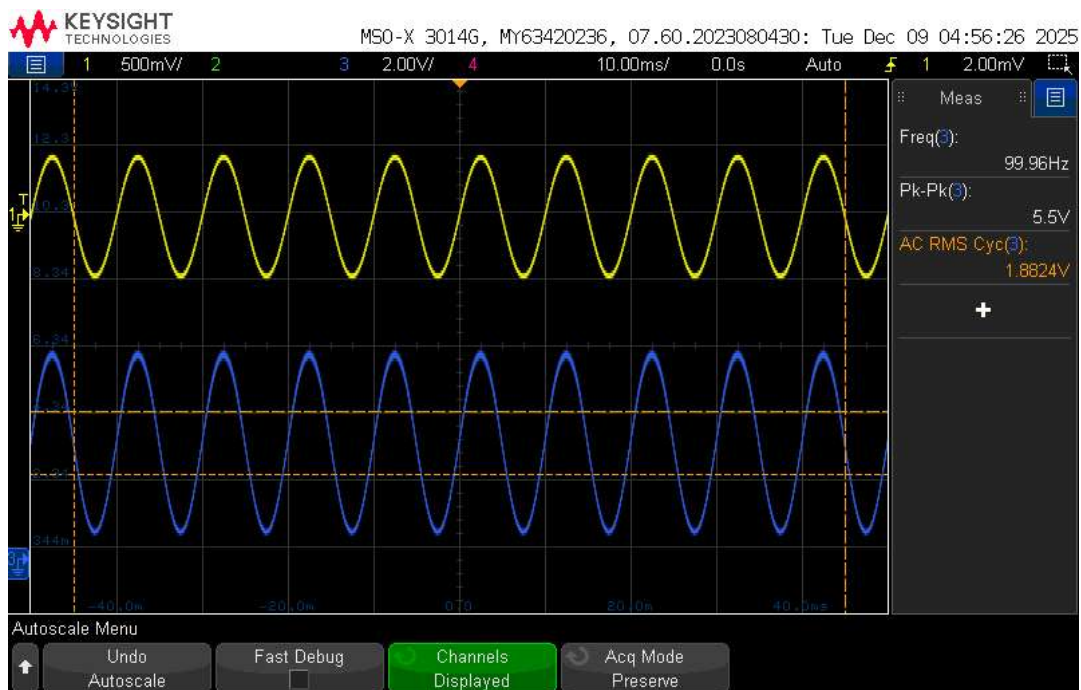


Figure 20: Oscilloscope at 100 Hz of power amplifiers

At 1000Hz:

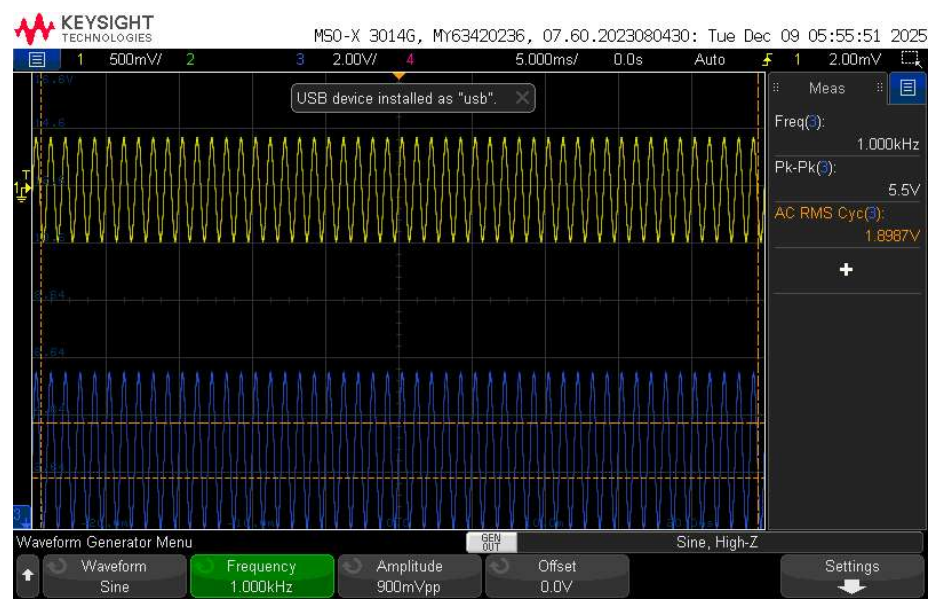


Figure 21: Oscilloscope at 1000Hz of power amplifiers.

At 10,000Hz:

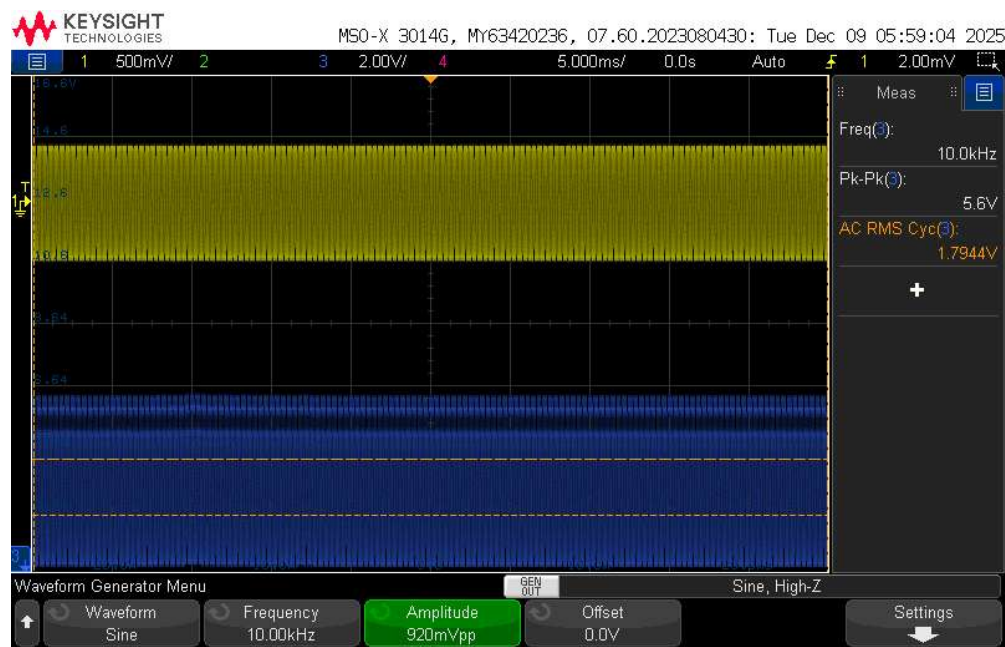


Figure 22: Oscilloscope at 10,000Hz of power amplifiers.

From the reading we can see at 100Hz it is 1.8824V, at 1k Hz its 1.8987 Hz and at 10k Hz it is 1.7944 Hz.

Calculation of output power from measured output voltage:

Using the RMS output voltage is related to power by:

$$P_{\text{out}} = \frac{V_{\text{rms}}^2}{R_L}$$

Where:

- P_{out} = output power (W)
- V_{rms} = measured RMS voltage across the load
- R_L = load resistance (Ω), here its 8 Ω

The V_{rms} value is 1.8824V substituting into the equation we get 442.93 mW which meets our requirements of greater than 400mW.

Plots demonstrating the low, medium and high-volume control:

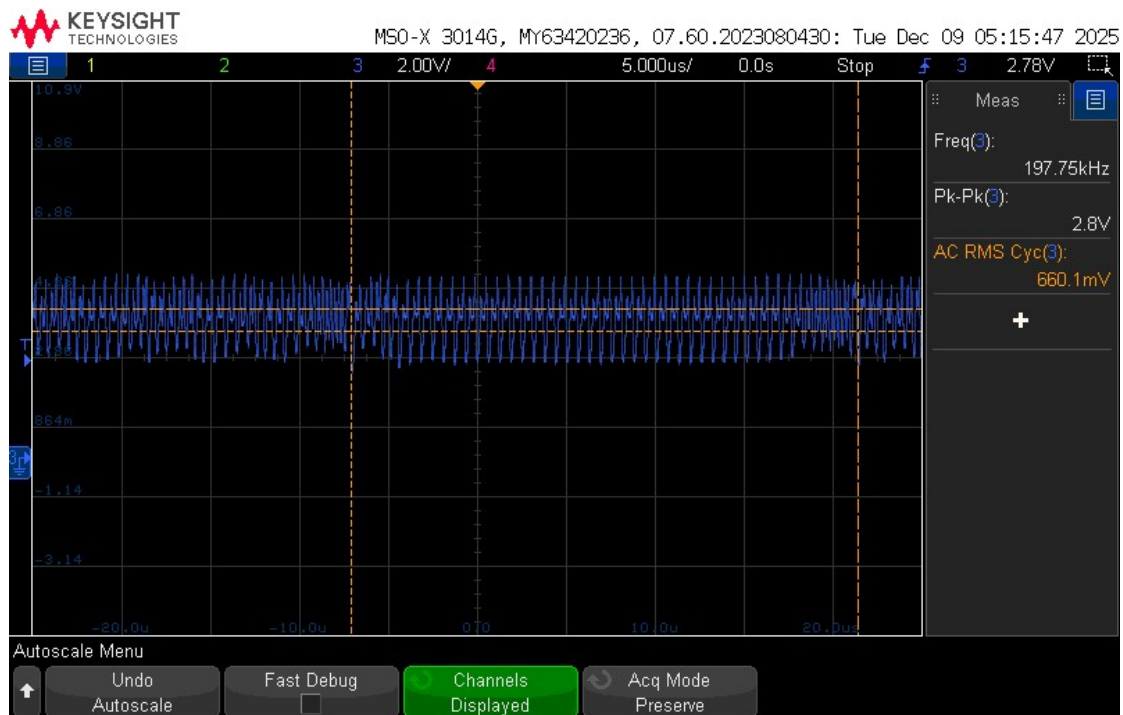


Figure 23: Oscilloscope at low audio.

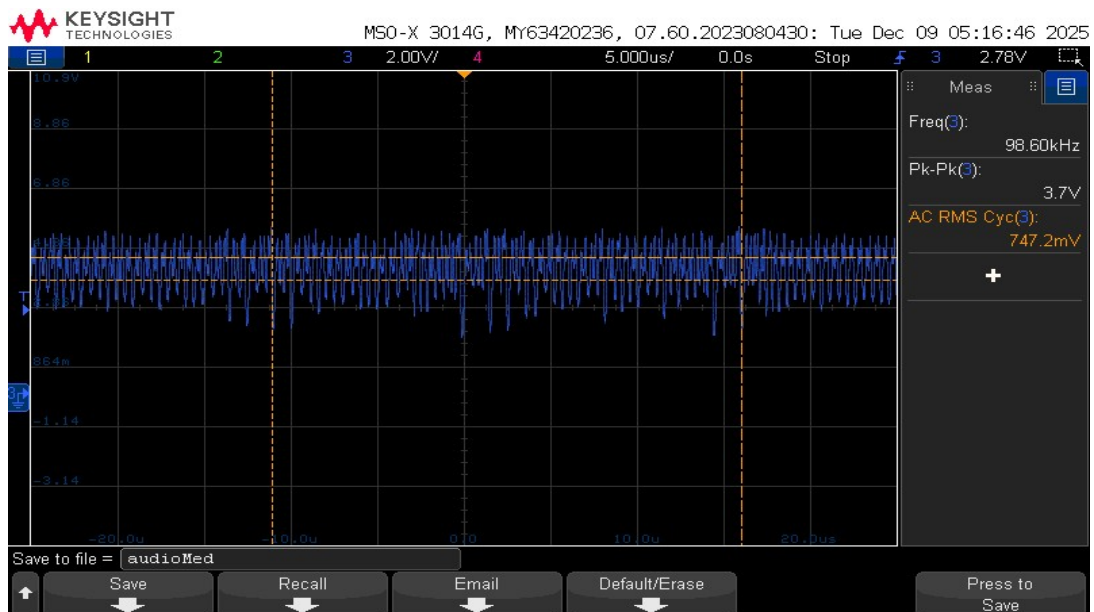


Figure 24: Oscilloscope at medium audio.

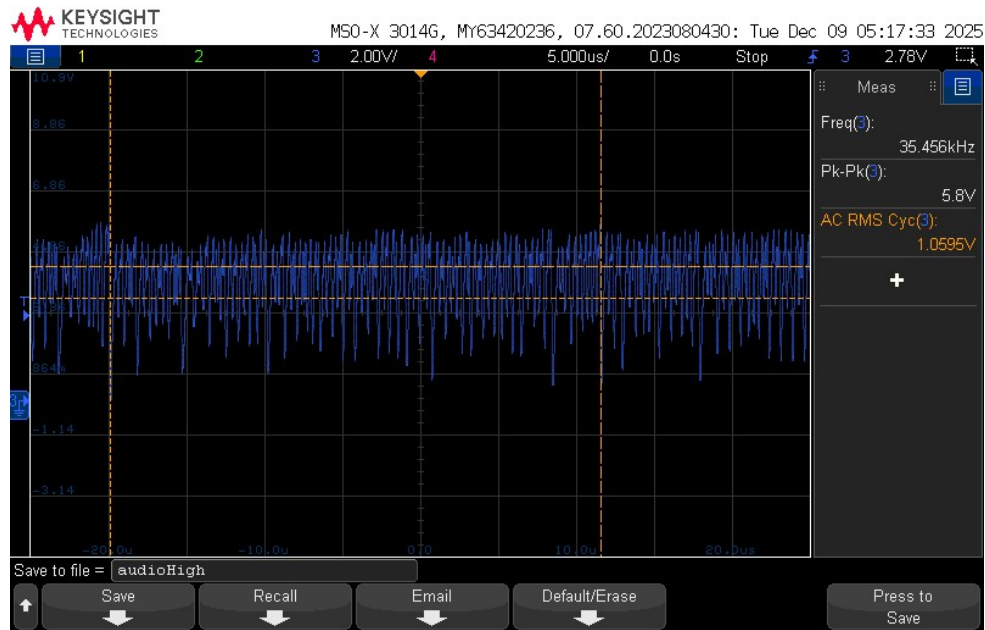


Figure 25: Oscilloscope at High audio.

Results Justification, Explanation, and Error/Tolerance Calculation:

The volume control (which is the potentiometer) acts as a variable voltage divider at the input stage of the amplifier. As a result, when the resistance is changed by turning the knob the input voltage supplied to the amplifier varies thus controlling the output amplitude.

This can be seen in the figures above where the V_{rms} for low audio is 660.1mV, for medium audio its 747.2 mV and for max it is 1.0595V increasing as you increase the resistance.

Error / Tolerance Calculation:

As far as error calculation is concerned the max V_{rms} at 10k Hz should be at 1V which well withing the range as the percentage error is 5.95% which is less than 10%.

Conclusion:

The objective of this project was to design and implement a 3-band audio equalizer with volume control including bass, mid, treble filters, and a voltage amplifier (Vamp) to drive an 8Ω load speaker. Then the system was designed to satisfy specifications such as -3 dB

cutoff frequencies, output power greater than 400 mW, maximum ripple of 15 mVrms, and proper volume control functionality such that the audio amplifier works properly.

From the overall findings we found out:

- The frequency of the equalizer was measured and within the range for bass (320 Hz), mid (320–3200 Hz), and treble (3200 Hz) bands within $\pm 10\%$ tolerance.
- Ripple measurements at the power supply output confirmed an RMS value of approximately 15 mVrms which is within the allowed tolerance.
- The amplifier output voltage consistently provided more than 400 mW of power across 100 Hz to 10 kHz using the equation.
- Volume control operation (low, medium, high) is accurate as when Vrms increased as the resistance increases.

Deviation from Expected Results:

When measuring the cutoff frequencies of certain filter bands, we got slightly different values than expected. Moreover, the Vrms for ripple calculations was also not the value we expected as we didn't get the frequency at 100 Hz when getting it.

Possible Reasons for Discrepancies:

Component tolerances: This could be due to resistive tolerances where the resistor and capacitor have their own resistance and as a result the values that we got for filter cutoff gain varied a little.

Measurement limitations: Such as probe loading where when using probes, it adds a small capacitance and resistance to the circuit, may give different values when measuring and coupling effect where when using AC coupling it introduces a high-pass filter effect, which can slightly shift phase.

References

- 1) Texas Instruments, *LM386 Low Voltage Audio Power Amplifier Datasheet*, SNAS545D, Rev. D, May 2004, revised Aug. 2023. [Online]. Available: <https://www.ti.com/lit/ds/symlink/lm386.pdf>
- 2) Texas Instruments, *LF356 JFET-Input Operational Amplifier — Product Details*, Texas Instruments. [Online]. Available: <https://www.ti.com/product/LF356>. Accessed: Dec. 12, 2025.
- 3) "Introduction to Operational Amplifiers," *All About Circuits*, [Online]. Available: <https://www.allaboutcircuits.com/textbook/semiconductors/chpt-8/introduction-operational-amplifiers/>.
- 4) Analog Devices, *Op Amp Applications Handbook – Section 1: Introduction to Operational Amplifiers*, [Online]. Available: <https://www.analog.com/media/en/training-seminars/design-handbooks/op-amp-applications/section1.pdf>.
- 5) MIT, *LF155 JFET-Input Operational Amplifier Datasheet*, [Online]. Available: <https://www.mit.edu/~6.301/LF155.pdf>.
- 6) "Low-Pass Active Filter," *Electronics Notes*, [Online]. Available: https://www.electronics-notes.com/articles/analogue_circuits/operational-amplifier-op-amp/low-pass-active-filter.php.
- 7) "Active Low-Pass Filter," *Circuit Digest*, [Online]. Available: <https://circuitdigest.com/tutorial/active-low-pass-filter>
- 8) "Introduction to Operational Amplifiers," *Microchip Technology*, [Online]. Available: <https://skills.microchip.com/introduction-to-operational-amplifiers/691922>
- 9) "Active Low-Pass Filter," *Electrical4U*, [Online]. Available: <https://www.electrical4u.com/active-low-pass-filter/>
- 10) R. Sam, "Op-amp based multiplier LTSpice simulation," *Electronics Stack Exchange*, Jun. 16, 2020. [Online]. Available: <https://electronics.stackexchange.com/questions/598688/op-amp-based-multiplier-ltspice-simulation>

